INSPIRE AI

STAND-UP

Share Your Day 5 Reflection



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Take turns sharing in your group

Use your records/orid.md to share a short reflection.

1. O · What happened? R · How did you feel?
2. I · What did you learn? D · What will you do next?
3. Let every member speak. Share only the feelings you are comfortable sharing.

Produce

One next action and any questions.

15 minutes · whole activity

Source activity: d06-s00

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REVIEW

Share Your Topic Concept Maps



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Explain one important connection

Each member shares their own graded concept map.

1. Divide the time equally, including feedback for each member.
2. Explain one important connection and how you checked its source.
3. Give specific feedback on accuracy, clarity or the source. Keep the submitted map and score unchanged.

Produce

Peer comments and questions to check later.

20 minutes · whole activity

Source activity: d06-s01

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PRACTICE

Explain a Rule and Run Its Test



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The operator needs a price or a reason

> As an operator, I want an explainable shipping price or a reason why a quote is unavailable.

Original request · US-01

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The allowed parcel inputs

US-01-AC01	Weight must be an integer from 1 to 20 kg. Zone must be local or remote. Service must be standard or express. Reject invalid inputs without a quote.

Selected criteria for US-01 · rules, not recorded test results

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Base price, extra weight and express service

US-01-AC02	The base price includes the first 2 kg. The local base price is 1000 cents. The remote base price is 2000 cents. Each kg above 2 adds 300 cents.
US-01-AC03	Express service adds 1500 cents to the weight and zone price. Standard service adds nothing.

Selected criteria for US-01 · rules, not recorded test results

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When remote express is unavailable

US-01-AC04	Remote express service is not available above 10 kg. Reject this combination without a quote. Remote standard service and local express service remain available up to 20 kg.

Selected criteria for US-01 · rules, not recorded test results

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A known rule example and its assertion

Given	Weight (kg) 1 Service standard	Zone local
When	Calculate the price.	
Then Expected result	Amount (cents) 1000	

Observation: Not recorded

Source example: US-01-E01

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Price examples to read and explain

Example	kg	Zone	Service	Expected result
US-01-E02	2	local	standard	1000 cents
US-01-E03	3	local	standard	1300 cents
US-01-E04	2	remote	standard	2000 cents
US-01-E05	2	local	express	2500 cents
US-01-E06	10	remote	express	5900 cents
US-01-E07	11	remote	standard	4700 cents
US-01-E08	20	local	express	7900 cents

Choose a price example and a rejection example. Trace each result to a rule, then locate its test.

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Rejection examples to read and explain

Example	kg	Zone	Service	Expected result
US-01-E09	11	remote	express	SERVICE_UNAVAILABLE · no quote
US-01-E10	0	local	standard	INVALID_INPUT · no quote
US-01-E11	21	local	standard	INVALID_INPUT · no quote

Choose a price example and a rejection example. Trace each result to a rule, then locate its test.

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More input rejection examples

Example	kg	Zone	Service	Expected result
US-01-E12	1.5	local	standard	INVALID_INPUT · no quote
US-01-E13	2	unknown	standard	INVALID_INPUT · no quote
US-01-E14	2	local	unknown	INVALID_INPUT · no quote

Choose a price example and a rejection example. Trace each result to a rule, then locate its test.

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Connect an example to a calculator test

Use the supplied quote rules, examples and calculator tests.

1. Follow one example from its input to a rule and assertion.
2. Run the supplied tests. Explain a different price or rejection.
3. Compare with a partner. Record a rule mismatch or a setup blocker.

Produce

Your rule explanation and actual test result, or a setup blocker.

20 minutes · whole activity

Source activity: d06-s02

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INSPIRE AI

LECTURE

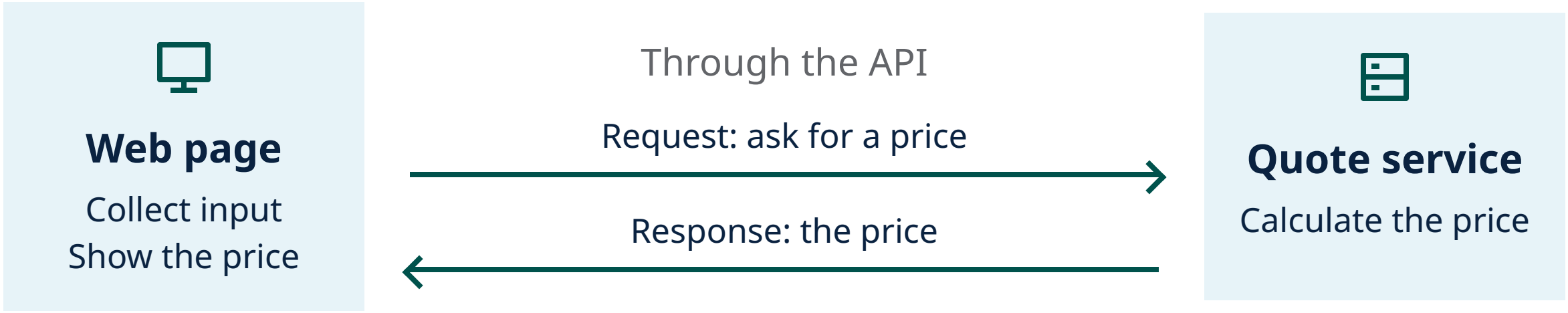
HTTP and RESTful APIs



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Programs use other programs

An API is an interface that lets one program use another program's functions.



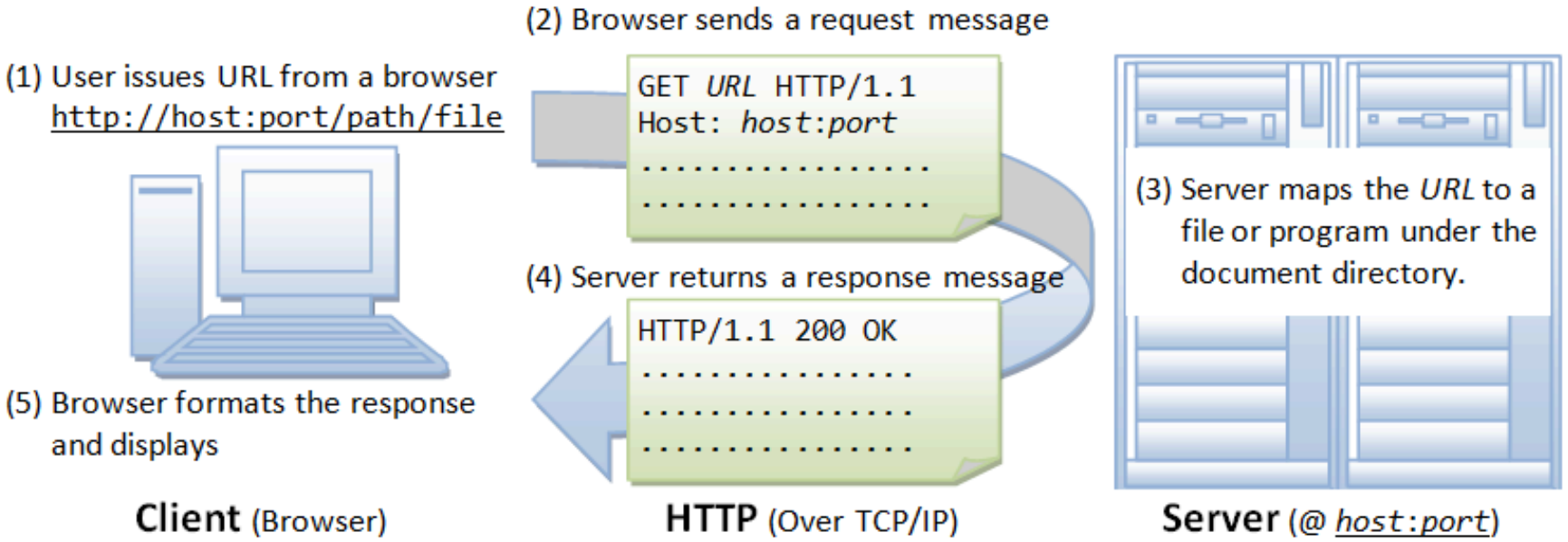
What can I ask for? What input? What comes back?

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HTTP: shared rules for messages

A protocol is a set of shared communication rules.

Client sends a request → **Server sends a response ←**



Responses can carry HTML pages, images, or JSON data.

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Read a message in three parts

Request →		← Response	
Method + target	What action? Which resource?	**Status code**	What was the result?
Headers	Extra information	**Headers**	Extra information
Body	Input data, when needed	**Body**	Returned data, when needed

For example, a Content-Type header tells us the body’s format.

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A URL tells us where to send a request

```
https://example.com/quotes/17
```

https	HTTP over an encrypted connection
example.com	Host: the server’s name
/quotes/17	Path: quote 17

Local example: http://localhost:8080 · localhost = this computer · 8080 = port

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REST organizes an API around resources

REST is a style for designing networked systems.

A resource is something we want to access or work with.

`/quotes`	The quote collection
`/quotes/17`	One quote, ID 17

Resource addresses + consistent operations

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The method and path describe the action

Request	Meaning in this API	Course role
`GET /quotes`	Read the list	Provided
`GET /quotes/17`	Read quote 17	Provided
`PUT /quotes/17`	Replace inputs and recalculate	Later learner transfer
`POST /quotes`	Create a quote	Demo, then learner practice

The path names the resource. The method names the action.

HTTP gives communication rules. REST gives design principles.

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JSON: field names and values

JSON is a text format for data.

One object: related fields inside { }

```
{
  "weight_kg": 2,
  "zone": "local"
}
```

Field names use double quotes.
Numbers do not need quotes.

Another JSON example

```
{
  "userId": 1,
  "id": 1,
  "title": "sunt",
  "body": "quia et suscipit"
}
```

Find the braces, a field name, and its value.

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JSON arrays hold a list of values

```
[{"id": 17}, {"id": 18}]
```

[ ] contains a list.
{ } contains one object.

Values: text, number, true / false, null, object, array.

`2` is a number. `"2"` is text.

`null` differs from a missing field.

```
[
  {
    "id": 1,
    "completed": false
  },
  {
    "id": 2,
    "completed": false
  }
]
```

Content-Type: application/json names the body format.

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A · Create a new quote

Example A · Expected behavior

Independent start: 0 saved quotes.

```
POST /quotes
Content-Type: application/json

{
  "weight_kg": 2,
  "zone": "local",
  "service": "express"
}
```

Request excerpt · the server generates the new quote ID.

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A · Read the successful response

Example A · Expected behavior

201 Created

```
Content-Type: application/json

{
  "weight_kg": 2,
  "zone": "local",
  "service": "express",
  "amount_cents": 2500
}
```

Response body excerpt · the response also includes a server-generated integer id.

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B · Check an unavailable service

Example B · Expected behavior

Independent start: 1 saved quote.

```
POST /quotes
Content-Type: application/json

{
  "weight_kg": 12,
  "zone": "remote",
  "service": "express"
}
```

Remote express service is unavailable above 10 kg.

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B · Reject the request without adding a quote

Example B · Expected behavior

422 Unprocessable Content

```
Content-Type: application/json

{
  "code": "SERVICE_UNAVAILABLE"
}
```

No new quote. Saved record count stays 1.

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Use the status code to find the result

2xx: success · 4xx: client error · 5xx: server error

200	Read or update succeeded
201	A quote was created
400	A field is missing or has the wrong type
404	The quote ID was not found
422	Input values or a service combination were rejected

These meanings follow this API’s contract. A missing weight differs from a weight of 0.

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Read the messages and explain them

A · Create a quote

```
POST /quotes
{ "weight_kg": 2,
  "zone": "local",
  "service": "express" }

201
{ "weight_kg": 2,
  "zone": "local",
  "service": "express",
  "amount_cents": 2500 }
```

B · Separate start

```
POST /quotes
{ "weight_kg": 12,
  "zone": "remote",
  "service": "express" }

422
{ "code": "SERVICE_UNAVAILABLE" }
```

Response excerpts · Remote express service is unavailable above 10 kg.

Identify: operation · input · success or rejection · returned data.

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Check your explanation

A · Create a quote

```
POST /quotes
{ "weight_kg": 2,
  "zone": "local",
  "service": "express" }

201
{ "weight_kg": 2,
  "zone": "local",
  "service": "express",
  "amount_cents": 2500 }
```

Created: 2500 cents. The server also returns a generated ID.

B · Separate start

```
POST /quotes
{ "weight_kg": 12,
  "zone": "remote",
  "service": "express" }

422
{ "code": "SERVICE_UNAVAILABLE" }
```

12 kg / remote / express: unavailable service. No new record.

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Read query parameters

```
https://example.com/items?limit=10#details
```

`?limit=10`	A query parameter; this API may define a ten-item limit
`#details`	A page fragment; not sent in the HTTP request

Multiple query parameters use &. URI identifies a resource; a URL also gives an access location and method.

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Four common operations

CRUD	Common HTTP method
Create	POST
Read	GET
Update	PUT replaces; PATCH describes partial changes
Delete	DELETE

The API contract defines the exact behavior.

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Read a nested structure

Resource path

```
/orders/17/items
```

Items belonging to order 17

Separate JSON example

```
{
  "quotes": [
    {"id": 17}
  ]
}
```

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Recognize more status codes

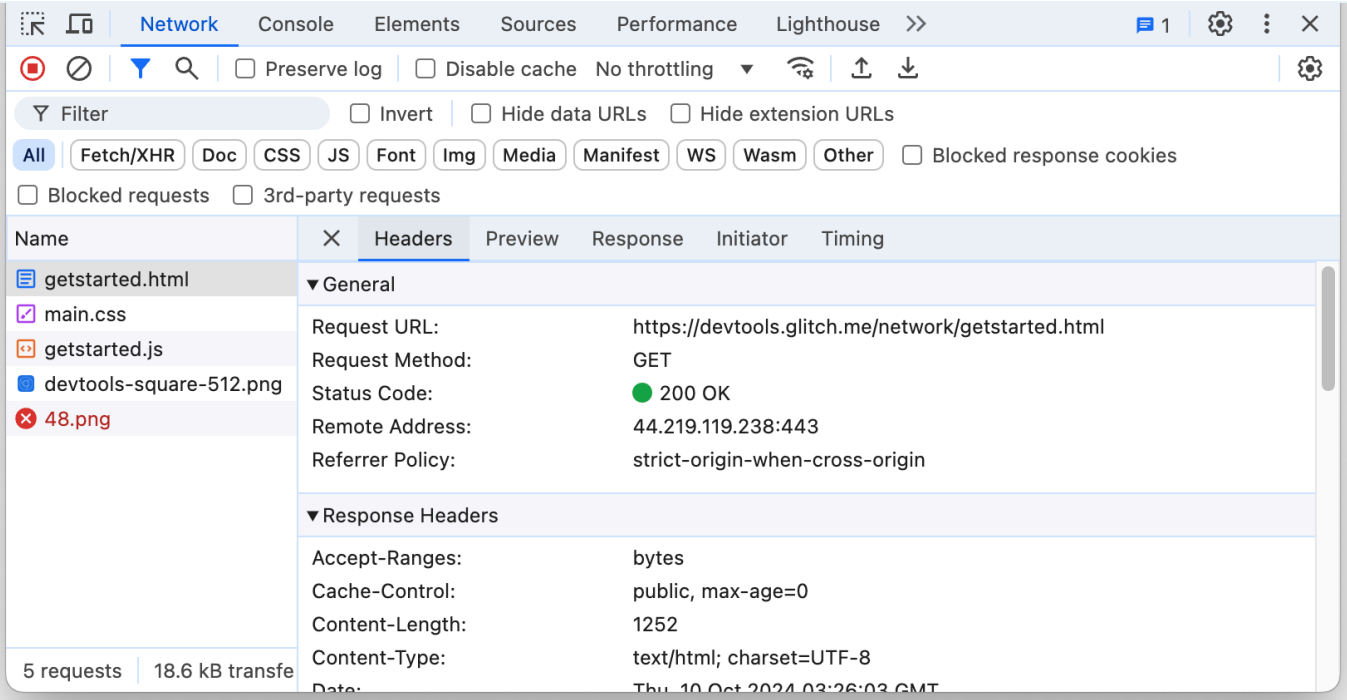
1xx / 3xx	Information / redirection
401	Valid authentication is needed
403	Access is refused
500	An unexpected server error

Recognize the group, then read the specific API response. You do not need to memorize every code.

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See requests and responses in the browser

Open Developer Tools → Network → reload the page.



1 · Select a request

Choose one row on the left.

2 · Read its details

Headers
URL, method, status

Response
Returned body

Payload
Request body, if present

Screenshot: Chrome DevTools documentation · This example loads an HTML page with GET.

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PRACTICE

Match Behavior to HTTP Messages



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Read two exchanges. Explain each part.

1. Use the two-page worksheet. Complete A on page 1, then B on page 2.
2. Write each label and its evidence beside the messages on the same page.
3. Answer the two questions in the spaces below each example.
4. Compare with a partner before opening the answer PDF.

These are expected messages from the case, not a captured server run.

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A · Read, label and explain

US-02-E05 · Saved quotes at start: 0 · Expected messages

Read the messages

```
POST /quotes
Content-Type: application/json
{ "weight_kg": 2,
  "zone": "local",
  "service": "express" }

201
Content-Type: application/json
{ "weight_kg": 2,
  "zone": "local",
  "service": "express",
  "amount_cents": 2500 }
```

Response excerpt: id is omitted. Explain its source below.

Write each answer + evidence

Resource / operation	
HTTP method	
Path	
Request input	
Response status	
Response body	

A1 · Is the target a collection or one saved quote? Why?

A2 · Who supplies the new ID? Where would it appear?

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B · Read, label and explain

US-02-E06 · Saved quotes at start: 1 · Expected messages

Read the messages

```
POST /quotes
Content-Type: application/json
{ "weight_kg": 12,
  "zone": "remote",
  "service": "express" }

422
Content-Type: application/json
{ "code": "SERVICE_UNAVAILABLE" }
```

Write each answer + evidence

Resource / operation	
HTTP method	
Path	
Request input	
Response status	
Response body	

Rule: remote express is unavailable above 10 kg. No new record; the collection must stay unchanged.

B1 · Does this response prove that the server is down?

B2 · Can the response body alone prove that no record was added?

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Compare the evidence, then revise

1. Take turns explaining A and B. Point to message text for each label.
2. Check request inputs against response fields. Discuss the generated ID.
3. Discuss B1 and B2. Separate the response from the stored state.
4. Keep one supported conclusion or one question. Revise your own notes after feedback.

> Keep your corrected table for the OpenAPI activity.

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Check A · Creating a quote

Resource / operation	Quote collection / create a quote
Method / path	POST /quotes
Request input	weight_kg: 2 · zone: local · service: express
Response status	201 · created
Response body	Input fields + amount_cents: 2500; also a generated integer id (omitted in the excerpt).

Source: US-02-E05. Method and path identify the attempted action; the response describes its result.

> A1 · POST targets the quote collection.
> A2 · The server generates the ID and returns it in the response.

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Check B · A business rejection

Resource / operation	Quote collection / create a quote
Method / path	POST /quotes
Request input	weight_kg: 12 · zone: remote · service: express
Response status	422 · business rejection
Response body	code: SERVICE_UNAVAILABLE

Source: US-02-E06. Method and path identify the attempted action; the response describes its result.

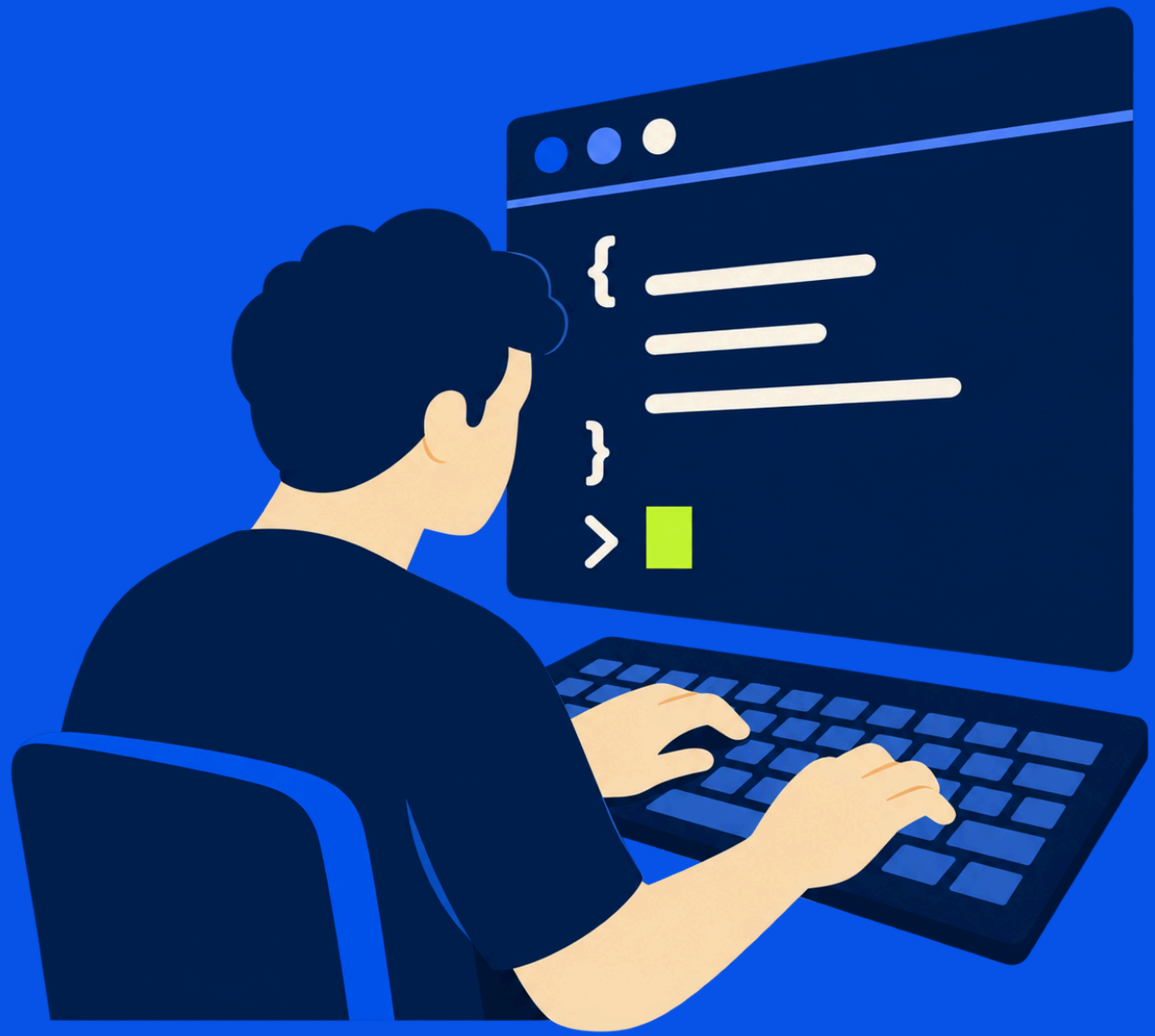
B1 · The response reports a rejected service combination.
B2 · Check stored records too. The error body alone is not proof.

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DEMO

Write the POST Contract in OpenAPI



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A quote that the operator can retrieve

> As an operator, I want to submit parcel details and receive a quote that I can retrieve later.

Original request · US-02

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Creation stores and returns a new quote

US-02-AC01	POST /quotes accepts weight_kg, zone, and service. Use the calculation rules in US-01. On success, store the inputs and amount_cents under a server-generated id. Return status 201 and the stored record.

Selected criteria for US-02 · rules, not recorded test results

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A rejected create adds no record

US-02-AC02	For invalid values, return status 422 with code INVALID_INPUT. For the unsupported service combination, return status 422 with code SERVICE_UNAVAILABLE. Neither rejection creates a record.

Selected criteria for US-02 · rules, not recorded test results

Today’s increment: SERVICE_UNAVAILABLE. INVALID_INPUT mapping stays in the later reference.

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Trainer example: an express quote

Given	Stored quotes at start 0	Weight (kg) 2
	Zone local	Service express
When	POST /quotes.	
Then Expected result	Status 201	Amount (cents) 2500
	ID generated; use the returned id in later checks	Stored quote count 1

Observation: Not recorded

Source example: US-02-E05

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Trainer example: unavailable service

Given	Stored quote count 1	Weight (kg) 12
	Zone remote	Service express
When	POST /quotes.	
Then Expected result	Status 422	Error code SERVICE_UNAVAILABLE
	Stored quote count 1	

Observation: Not recorded

Source example: US-02-E06

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Record the create behavior before coding

Describe POST /quotes in the supplied OpenAPI starter.

1. Use the three input fields and a server-generated ID.
2. Add one success response and one rejection response, with example values.
3. Validate the YAML. Compare both responses with the agreed behavior.

Produce

A validated POST operation to use as a reference.

15 minutes · whole activity

Source activity: d06-s05

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PRACTICE

Write Your POST Contract



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Your example: a standard quote

Given	Stored quotes at start 0	Weight (kg) 3
	Zone local	Service standard
When	POST /quotes.	
Then Expected result	Status 201	Amount (cents) 1300
	ID generated; use the returned id in later checks	Stored quote count 1

Observation: Not recorded

Source example: US-02-E01

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Your example: rejection without a new quote

Given	Stored quote count 1	Weight (kg) 11
	Zone remote	Service express
When	POST /quotes.	
Then Expected result	Status 422	Error code SERVICE_UNAVAILABLE
	Stored quote count 1	

Observation: Not recorded

Source example: US-02-E02

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Describe POST /quotes in your own file

Use CREATE-QUOTE examples US-02-E01 and US-02-E02.

1. Use the starter to add the method, request and example inputs.
2. Add success and rejection responses. Validate the YAML.
3. Use the final 10 minutes to check both files with a partner. Explain one choice.

Produce

Your OpenAPI file, peer comments and any unresolved question.

25 minutes · whole activity

Source activity: d06-s06

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SHOWCASE

Compare and Revise Your Contracts



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Connect a contract choice to its example

Check the meaning of the contract as well as its syntax.

1. Selected presenters show one choice and the example behind it.
2. Compare the feedback with your own success and rejection responses.
3. Correct your file. Record any rule that still needs confirmation.

Produce

Your revised contract, ready for API tests.

10 minutes · whole activity

Source activity: d06-s07

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INSPIRE AI

LECTURE

From HTTP Requests to Java Objects



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Reading returns the saved data

US-02-AC03	GET /quotes/{id} returns status 200 and the saved record for an existing id. GET /quotes returns saved records in ascending id order. Reading does not change data.

Selected criteria for US-02 · rules, not recorded test results

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The existing quote and its GET response

GET /quotes/17.

Given

Saved ID 17

Saved quote

Weight (kg)	Zone	Service	Amount (cents)
3	local	standard	1300

Then · Expected result

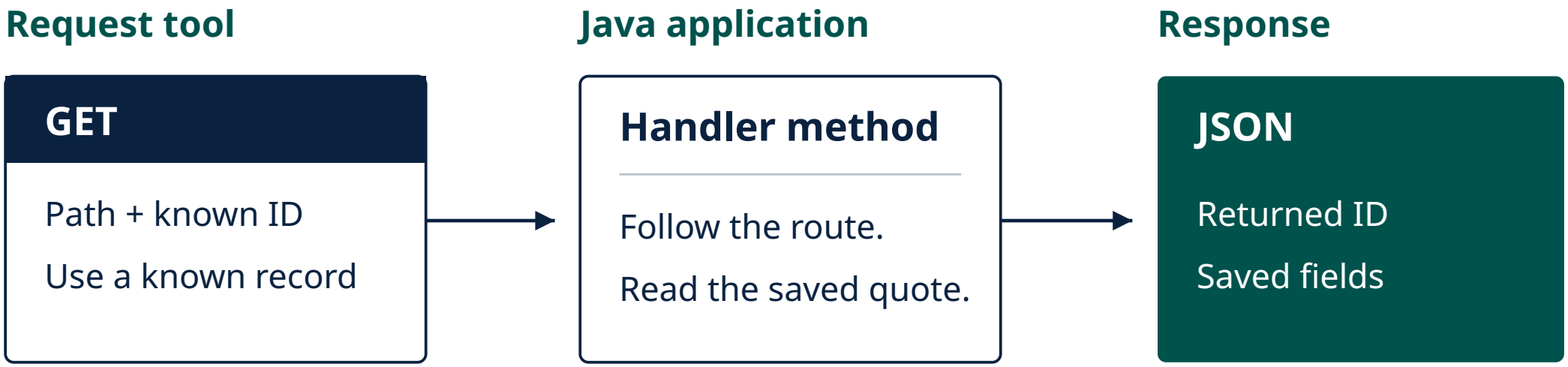
Status	ID	Weight (kg)	Zone	Service	Amount (cents)
200	17	3	local	standard	1300

US-02-E04 · Specification expectation · Actual result: not recorded

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Start with a request that already works

Read one existing quote in the supplied application.



Keep all three visible as you trace the request.

Live walkthrough · This read is supplied. PUT and POST are the next tasks.

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A route connects HTTP to a Java method

Method + path	Select the Controller method.
Path value	Becomes a Java argument, such as the record ID.
Returned object	Becomes JSON in the response.

An annotation is a marker that the framework reads. Point to these connections in the IDE.

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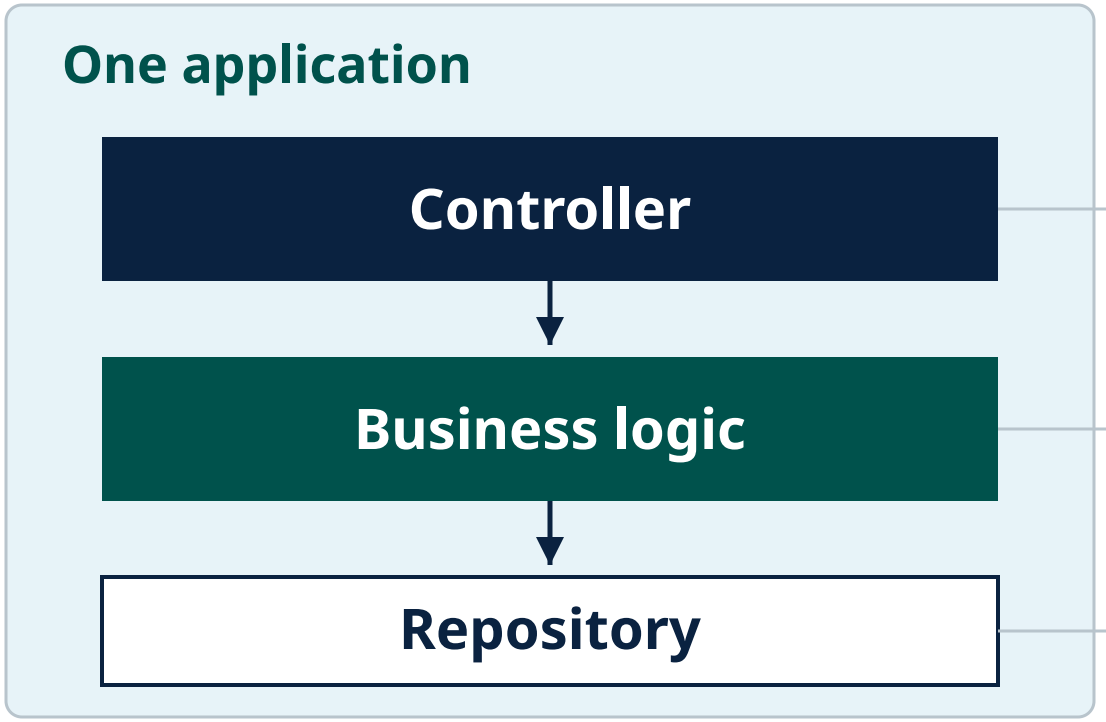
Three names, three useful roles

Spring	Creates and connects configured objects.
Spring MVC	Handles web requests and responses.
Spring Boot	Helps configure and start the application.

Our code still defines the business behavior.

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Separate three kinds of work



HTTP input and response
Read inputs. Return status and fields.

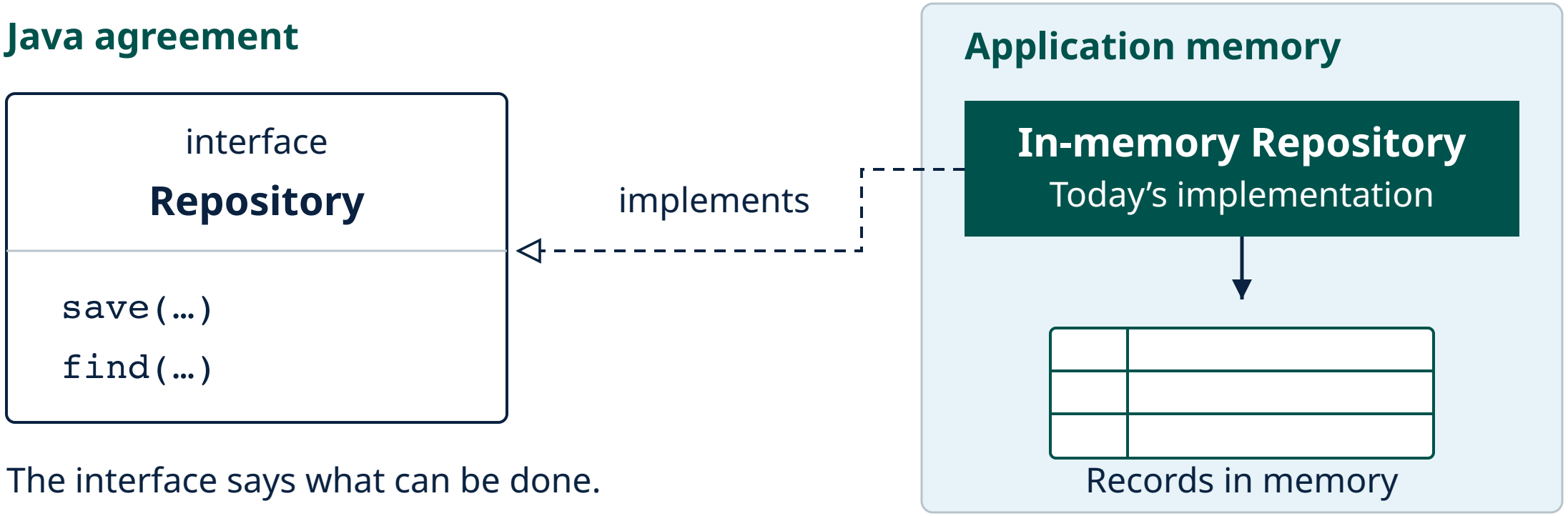
Rules and coordination
Apply rules. Coordinate the operation.

Record access
Save and find records in memory today.

One application, different responsibilities. Reading a saved quote does not require recalculating it.

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A Repository does not have to use a database

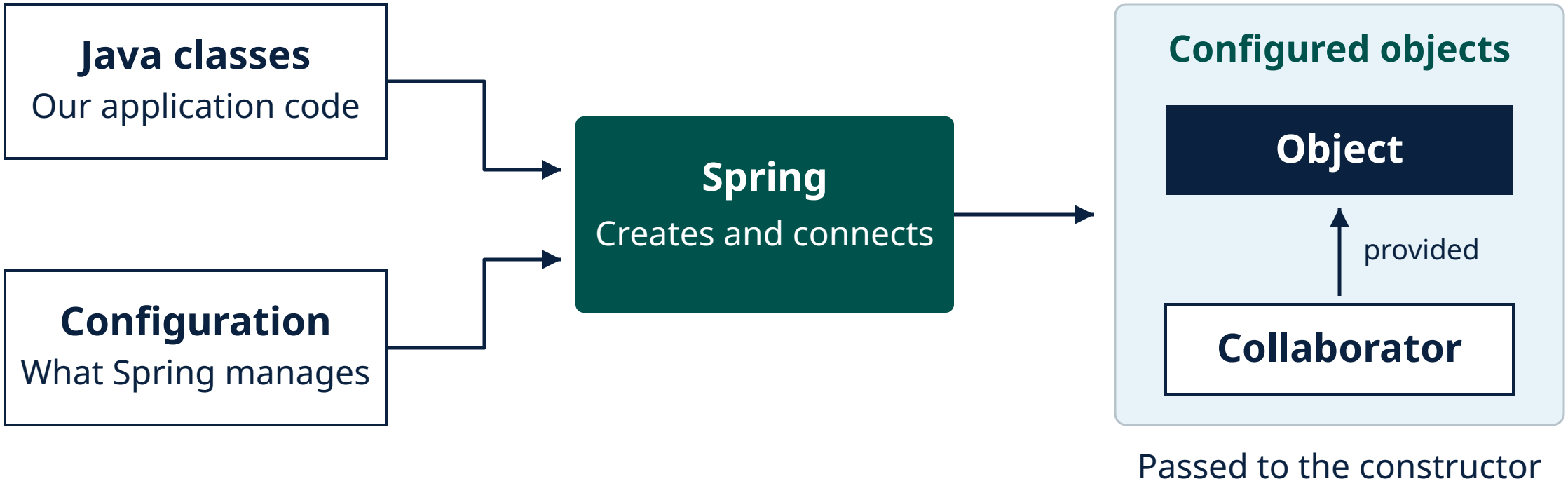


The interface says what can be done.

Day 7 changes the storage implementation. No database is connected in today’s path.

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Who gives an object its collaborators?



> Dependency injection: provide the objects a class needs.

Adapted from Spring's container overview. In the IDE: find the parameter type and the supplied object.

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Point to the code, then explain the path

1. Where does the request enter?
2. Where is the record read, and where does it live?
3. Which concrete object is passed to this constructor?

> Does this successful read prove that a database row exists?

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A successful read tells us about today’s path

Entry	The route selects a Controller method.
Storage	The current Repository reads application memory.
Collaboration	The constructor receives the configured object.

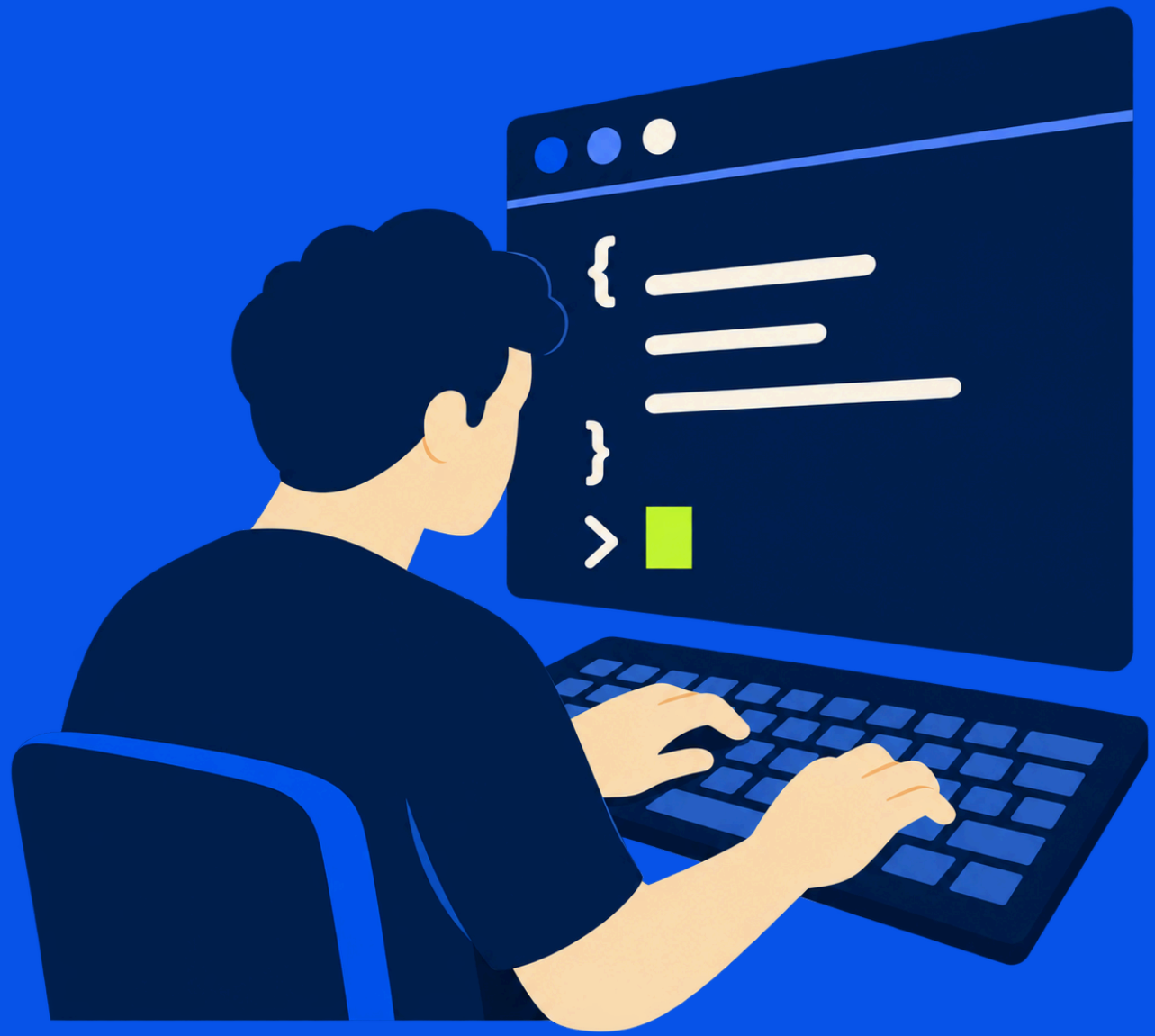
Next: let a test send that same GET and check its response.

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DEMO

Send the Same GET with MockMvc



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The test entry changes what we can check

Calculator test

Call a Java method. Check its returned price.

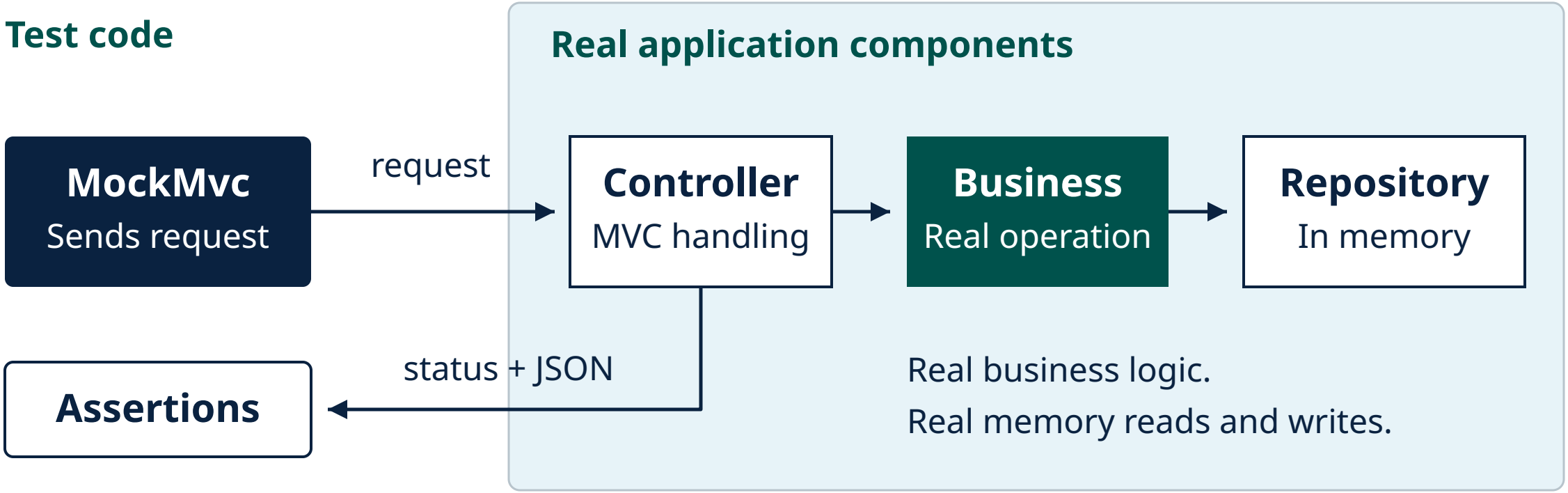
API test

Send a request through MVC. Check status and response fields.

> A correct calculation can still be returned incorrectly.

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MockMvc sends the request; our code runs



Real MVC, business objects and in-memory records. No listening port or database.

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The same GET, with independent test data

Manual GET

Read one of the records supplied at application startup.

MockMvc GET

Reset, seed directly, send GET and assert the response.

> Each test prepares its own starting state.

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The GET example becomes test assertions

GET /quotes/17.

Given

Saved ID	17

Saved quote

Weight (kg)	Zone	Service	Amount (cents)
3	local	standard	1300

Then · Expected result

Status	ID	Weight (kg)	Zone	Service	Amount (cents)
200	17	3	local	standard	1300

US-02-E04 · Specification expectation · Actual result: not recorded

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Run the supplied GET test

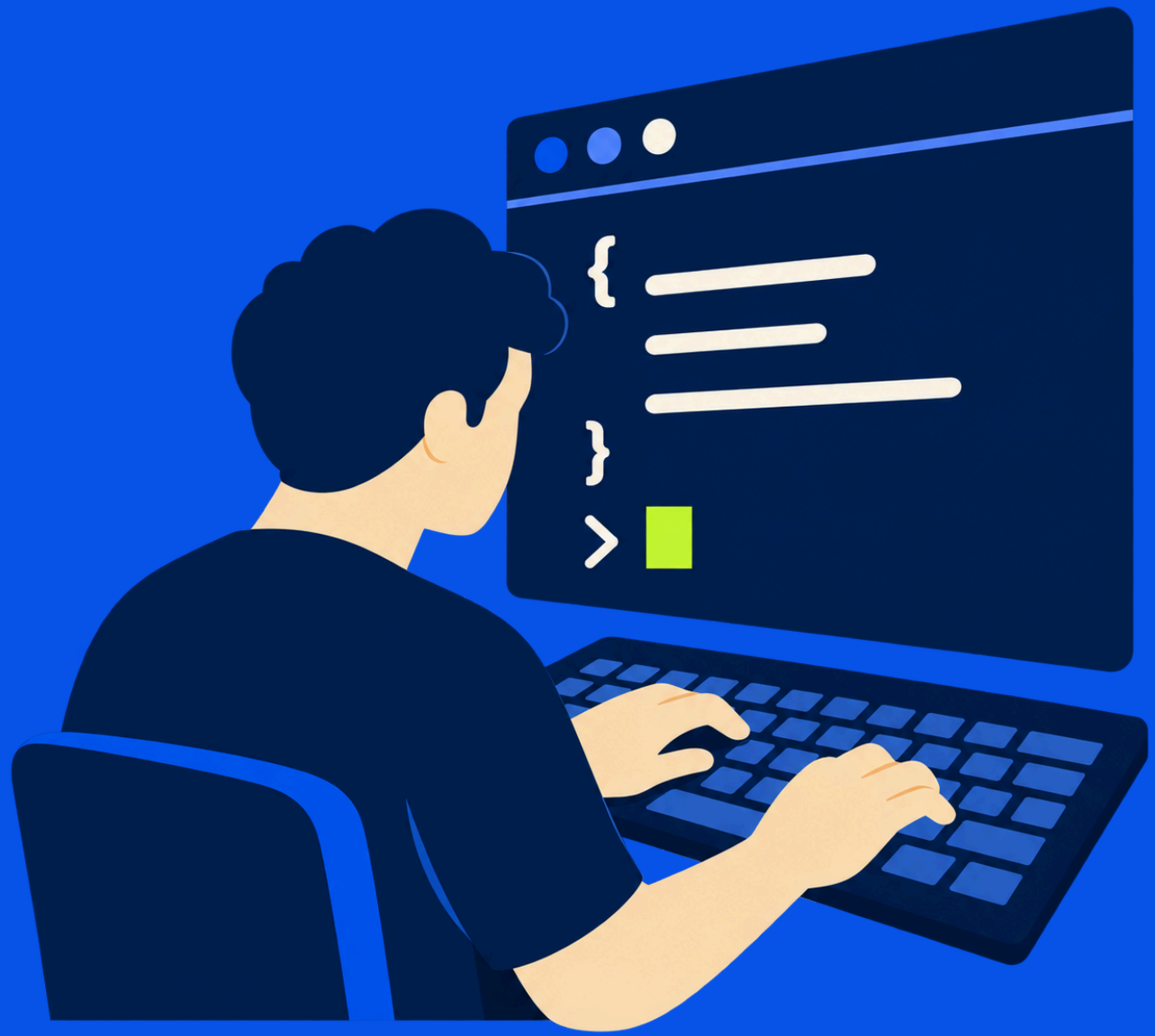
1. Point to the request and its assertions.
2. Explain where the starting record comes from.
3. Run the test and inspect the result.

> The supplied GET should pass. Next: a new POST behavior.

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DEMO

Write the POST Test Before Coding



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The example defines the assertions

Given	Stored quotes at start 0	Weight (kg) 2
	Zone local	Service express
When	POST /quotes.	
Then Expected result	Status 201	Amount (cents) 2500
	ID generated; use the returned id in later checks	Stored quote count 1

Observation: Not recorded

Source example: US-02-E05

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Arrange, act, assert, then run

1. Reset the real in-memory repository.
2. Send JSON with Content-Type: application/json.
3. Check status, generated identity and business fields.
4. Run before changing product code.

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Explain the result before changing code

1. Did the test compile and start?
2. Did it use the intended request and starting state?
3. Which expected behavior is missing?

Check the route and setup before treating a failure as evidence.

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PRACTICE

Write Your POST Success Test



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The same operation, your own example

Given	Stored quotes at start 0	Weight (kg) 3
	Zone local	Service standard
When	POST /quotes.	
Then Expected result	Status 201	Amount (cents) 1300
	ID generated; use the returned id in later checks	Stored quote count 1

Observation: Not recorded

Source example: US-02-E01

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Write, run and save your first result

1. Trace each expected field to your example.
2. Write and run the POST success test.
3. Explain the result and fix setup errors.
4. Save your test and its result.

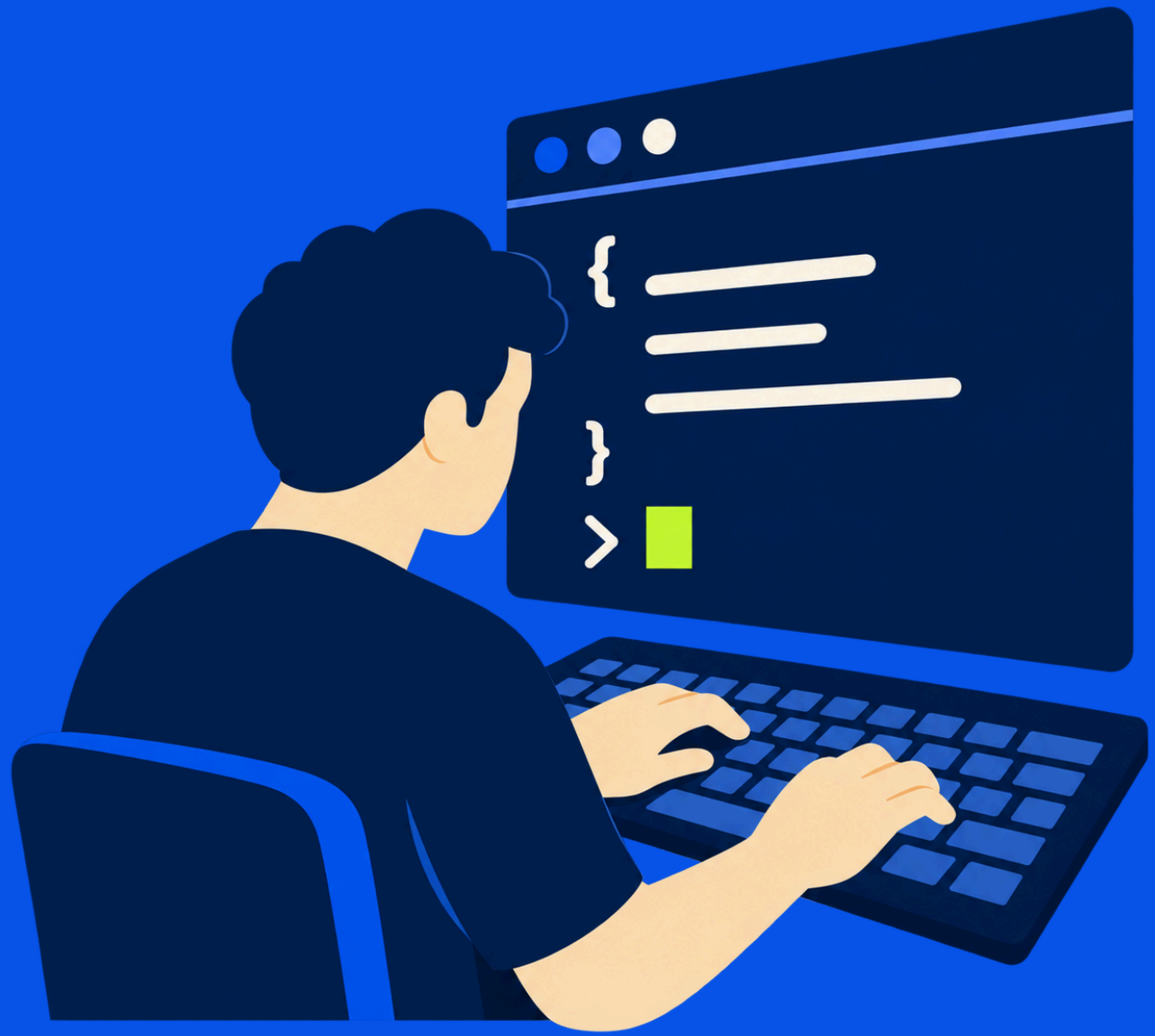
Keep rejection on the test list until success passes.

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DEMO

Complete the POST Success and Rejection Cycles



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The same success example guides the code

Given	Stored quotes at start 0	Weight (kg) 2
	Zone local	Service express
When	POST /quotes.	
Then Expected result	Status 201	Amount (cents) 2500
	ID generated; use the returned id in later checks	Stored quote count 1

Observation: Not recorded

Source example: US-02-E05

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Connect the behavior the success test needs



The expected response stays the same throughout this change.

Return the generated quote with 201. Rerun the same success test.

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Check rejection and the stored records

Given	Stored quote count 1	Weight (kg) 12
	Zone remote	Service express
When	POST /quotes.	
Then Expected result	Status 422	Error code SERVICE_UNAVAILABLE
	Stored quote count 1	

Observation: Not recorded

Source example: US-02-E06

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PRACTICE

Complete Your POST Behavior



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Your saved success test guides the code

Given	Stored quotes at start 0	Weight (kg) 3
	Zone local	Service standard
When	POST /quotes.	
Then Expected result	Status 201	Amount (cents) 1300
	ID generated; use the returned id in later checks	Stored quote count 1

Observation: Not recorded

Source example: US-02-E01

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Finish success, then check rejection

1. Make your saved success test pass.
2. Add the service-rejection test with independent data.
3. Explain its result, then implement any missing mapping.
4. Rerun success and rule tests; correct one finding.

> Keep your passing POST. You will extend this same project.

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After success: your rejection example

Given	Stored quote count 1	Weight (kg) 11
	Zone remote	Service express
When	POST /quotes.	
Then Expected result	Status 422	Error code SERVICE_UNAVAILABLE
	Stored quote count 1	

Observation: Not recorded

Source example: US-02-E02

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LECTURE

Adapt the Contract and Setup for an Update



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The operator replaces an existing quote

> As an operator, I want to recalculate and replace an existing quote.

Original request · US-03

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Replacement keeps the ID and protects old data

US-03-AC01	PUT /quotes/{id} replaces the inputs and calculated amount for an existing id using US-01. Return status 200 and the updated record with the same id.
US-03-AC02	Use the rejection codes and input shape from US-02. A rejected update leaves the stored record unchanged. An unknown id returns status 404.

Selected criteria for US-03 · rules, not recorded test results

Classroom rejection: SERVICE_UNAVAILABLE. Retain all old fields on rejection.

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An update changes three test decisions

Decision	POST	PUT
Target	Collection path	Existing ID in the path
Identity	Server generates it	The ID stays the same
Success	201 Created	200 OK

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Prepare the existing target directly

Arrange	Reset, then seed the update target.
Success	Replace its inputs and amount. Keep its ID.
Rejection	Keep every stored field unchanged.

Locate the supplied PUT outline and seed helper. Keep this test independent of POST.

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An update returns the same ID

Given	Existing ID 17	Weight (kg) 2
	Zone local	Service express
When	PUT /quotes/17.	
Then Expected result	Status 200	ID 17
	Amount (cents) 2500	

Observation: Not recorded

Source example: US-03-E01

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Rejection preserves the stored quote

Given	Existing ID 17 Weight (kg) 11 Service express	Old amount (cents) 1300 Zone remote
When	PUT /quotes/17.	
Then Expected result	Status 422 Stored amount (cents) 1300	Error code SERVICE_UNAVAILABLE

Observation: Not recorded

Source example: US-03-E03

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PRACTICE

Test and Implement the Update



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Your PUT success assertions

Given	Existing ID 17	Weight (kg) 2
	Zone local	Service express
When	PUT /quotes/17.	
Then Expected result	Status 200	ID 17
	Amount (cents) 2500	

Observation: Not recorded

Source example: US-03-E01

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Extend your contract, then write the test

1. Add PUT to your OpenAPI file from the selected examples.
2. Seed the target directly and write the success test.
3. Check 200, the same ID and expected business fields.
4. Run, explain and save the result.

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Make the update pass and preserve POST

1. Use the saved result to make the smallest update.
2. Add the service-rejection check; compare the old record.
3. Rerun POST, PUT and rule tests.
4. Review a peer’s evidence, correct and rerun.

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Your PUT rejection and stored-state check

Given	Existing ID 17	Old amount (cents) 1300
	Weight (kg) 11	Zone remote
	Service express	
When	PUT /quotes/17.	
Then Expected result	Status 422	Error code SERVICE_UNAVAILABLE
	Stored amount (cents) 1300	

Observation: Not recorded

Source example: US-03-E03

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A new rejection test may already pass

Inspect the evidence

Confirm the test reaches PUT and checks the entire old record.

Explain the result

The existing shared error handler may already provide this response.

> Keep useful passing coverage. Make only a needed change.

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BREAK

Save Your POST and PUT Results

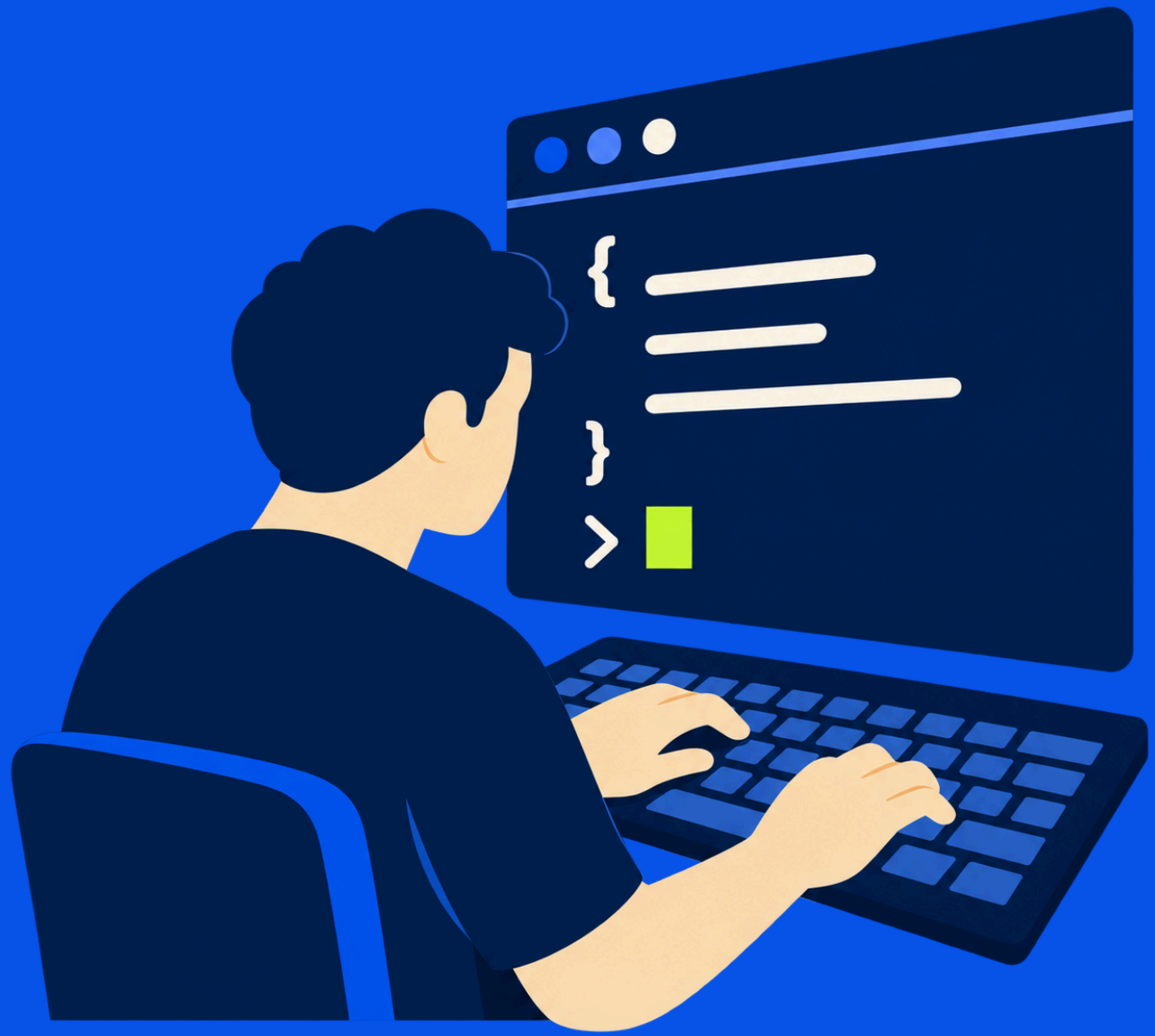


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DEMO

Delete One Quote with an API Test



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The operator removes one unwanted quote

> As an operator, I want to delete quotes I no longer need while keeping other quotes.

Original request · US-08

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Deletion removes only the selected quote

US-08-AC01	DELETE /quotes/{id} removes an existing quote and returns status 204 with an empty body. A later GET for that id returns 404. All other quotes remain unchanged.
US-08-AC02	DELETE /quotes/{id} for an unknown id returns status 404 and leaves the entire collection unchanged. It does not calculate prices or check service availability.

Selected criteria for US-08 · rules, not recorded test results

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Delete the target. Preserve the other quote.

Existing ID

204, empty body. GET of the deleted target returns 404.

Unknown ID

404. Every stored record stays unchanged.

DELETE /quotes/{id} · No request body · No price calculation

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Trainer example: two independent records

DELETE /quotes/17 from this independently seeded state.

Given

Stored quotes

ID	Weight (kg)	Zone	Service	Amount (cents)
17	3	local	standard	1300
18	2	local	express	2500

US-08-E01 · Specification expectation · Actual result: not recorded

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Trainer example: the target is gone

DELETE /quotes/17 from this independently seeded state.

Then · Expected result

Status	204
Response body	Empty body
GET of deleted ID	404

Stored quotes

ID	Weight (kg)	Zone	Service	Amount (cents)
18	2	local	express	2500

US-08-E01 · Specification expectation · Actual result: not recorded

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Write the test before the delete method

1. Reset, then seed the target and a second quote directly.
2. Send DELETE; expect 204 and an empty body.
3. GET the target; expect 404.
4. Compare all fields of the remaining records, then run.

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Remove only the requested record

1. Check that the target exists.
2. Remove that ID from the in-memory repository.
3. Return 204 with no body.
4. Rerun the same test, then the existing POST and GET tests.

> Why is checking only 204 insufficient?

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Unknown ID: the collection stays unchanged

DELETE /quotes/999 from this independently seeded state.

Given

Stored quotes

ID	Weight (kg)	Zone	Service	Amount (cents)
17	3	local	standard	1300
18	2	local	express	2500

Then · Expected result

Status 404

Stored quotes

ID	Weight (kg)	Zone	Service	Amount (cents)
17	3	local	standard	1300
18	2	local	express	2500

US-08-E02 · Specification expectation · Actual result: not recorded

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PRACTICE

Test and Implement DELETE



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Your records: delete 27 and preserve 28

DELETE /quotes/27 from this independently seeded state.

Given

Stored quotes

ID	Weight (kg)	Zone	Service	Amount (cents)
27	3	local	standard	1300
28	2	local	express	2500

US-08-E03 · Specification expectation · Actual result: not recorded

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Your expected response and remaining record

DELETE /quotes/27 from this independently seeded state.

Then · Expected result

Status	204
Response body	Empty body
GET of deleted ID	404

Stored quotes

ID	Weight (kg)	Zone	Service	Amount (cents)
28	2	local	express	2500

US-08-E03 · Specification expectation · Actual result: not recorded

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Use your own pair of records

1. Add DELETE to your OpenAPI contract.
2. Seed both records from your assigned example.
3. Test the empty 204, missing target and unchanged survivor.
4. Run, explain the result and implement the missing behavior.

Continue your POST / PUT project. Use the outline and syntax reference.

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An unknown ID must leave everything unchanged

Arrange and act

Reset, seed both records, snapshot the collection and DELETE an unknown ID.

Assert

Expect 404. Compare the entire collection before and after.

> Already passing? Explain which existing behavior supports it.

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Your unknown-ID example starts fresh

DELETE /quotes/999 from this independently seeded state.

Given

Stored quotes

ID	Weight (kg)	Zone	Service	Amount (cents)
27	3	local	standard	1300
28	2	local	express	2500

Then · Expected result

Status 404

Stored quotes

ID	Weight (kg)	Zone	Service	Amount (cents)
27	3	local	standard	1300
28	2	local	express	2500

US-08-E04 · Specification expectation · Actual result: not recorded

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Rerun, compare and save your evidence

1. Validate OpenAPI; rerun POST, PUT, DELETE and rule tests.
2. Review independent setup, empty body and preserved records with a peer.
3. Correct one actual finding and rerun.
4. Save your results and any unfinished work for the showcase.

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SHOWCASE

Show Your Contract, Code and Tests



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Show the evidence for one behavior

Trace one agreed example through your work.

1. Show the example, contract, first test result, code change and latest checks.
2. Explain what your API test checks and what the calculator test checks.
3. Record one correction or evidence gap in your own work.

Produce

Your revision notes and remaining questions.

25 minutes · whole activity

Source activity: d06-s18

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HOMEWORK

Build and Test a CRUD API



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Templates keep inputs for future quotes

> As an operator, I want to create, read, update and delete templates of common parcel inputs so that I can maintain inputs for future quotes.

Original request · US-07

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An independent resource with five fields

US-07-AC01	A quote template contains id, name, weight_kg, zone, and service. Store templates in a separate in-memory repository for the current process. Responses contain these five fields and no amount_cents. Template management does not calculate prices, check service availability, create quotes, or apply a template to a quote.

Selected criteria for US-07 · rules, not recorded test results

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Continue in your Day 6 homework repository

Your Day 6 classroom project in your personal homework repository

Before class, your personal Day 6 homework repository is initialized from the approved parcel-quote main snapshot. Clone your assigned repository and use it throughout the day. For homework, continue from your own classroom quote API work in that same repository. No template scaffold is supplied. Use the quote code and tests as references to build the template resource yourself, preserving your classroom code and commit history. Run ./mvnw test on macOS or Linux, or .\mvnw.cmd test in Windows PowerShell, before editing. Record any unfinished classroom behavior or failing baseline before starting the template task.

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Homework: quote template CRUD

Independently describe, implement and test create, read, replacement and delete for one in-memory resource in the existing case.

1. Check Your Classroom Baseline and the Template Examples
2. Write the API Contract
3. Test and Implement Create and Read
4. Test and Implement Update and Delete
5. Run Regression and Submit

Produce

OpenAPI contract

Runnable project and API tests

TDD and regression results

60 minutes · whole activity

Source activity: d06-hw

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